NYC Public School SAT Performance

Exploratory Data Analysis and Borough-Level Comparison

This report examines school-level averages for the legacy SAT Math, Critical Reading and Writing sections. The three section averages are summed to create a school-level comparison metric.

Dataset source: Kaggle — NYC Public School SAT Performance. The Kaggle upload date is not necessarily the test year; the exact testing year for this copy is not established in this repository.

Unit of analysis: one school per row. SAT values are school-level averages, not individual student scores.

Score-format caution: these are legacy three-section scores (600–2,400 combined), not current Digital SAT totals (400–1,600).

Dataset overview and key findings

The file contains 375 school records across 5 boroughs. All 375 records have complete Math, Reading and Writing averages. 20 records have no reported participation percentage.

Math screening: 10 schools meet the project-defined 640/800 threshold (80% of the section maximum). This is not an official College Board proficiency benchmark.

Top combined school-level score: Stuyvesant High School (2,144 out of 2,400).

Highest borough mean: Staten Island (1,439.00; n=10). **Largest borough sample standard deviation:** Manhattan (230.29; n=89). Standard deviation measures spread, not achievement.

Potential IQR outliers: 27. These are flags for investigation, not automatic data errors or grounds for deletion.

Participation association: Pearson r=0.604 and Spearman rho=0.606 across 355 schools with reported participation. This is association, not evidence that participation causes higher scores.

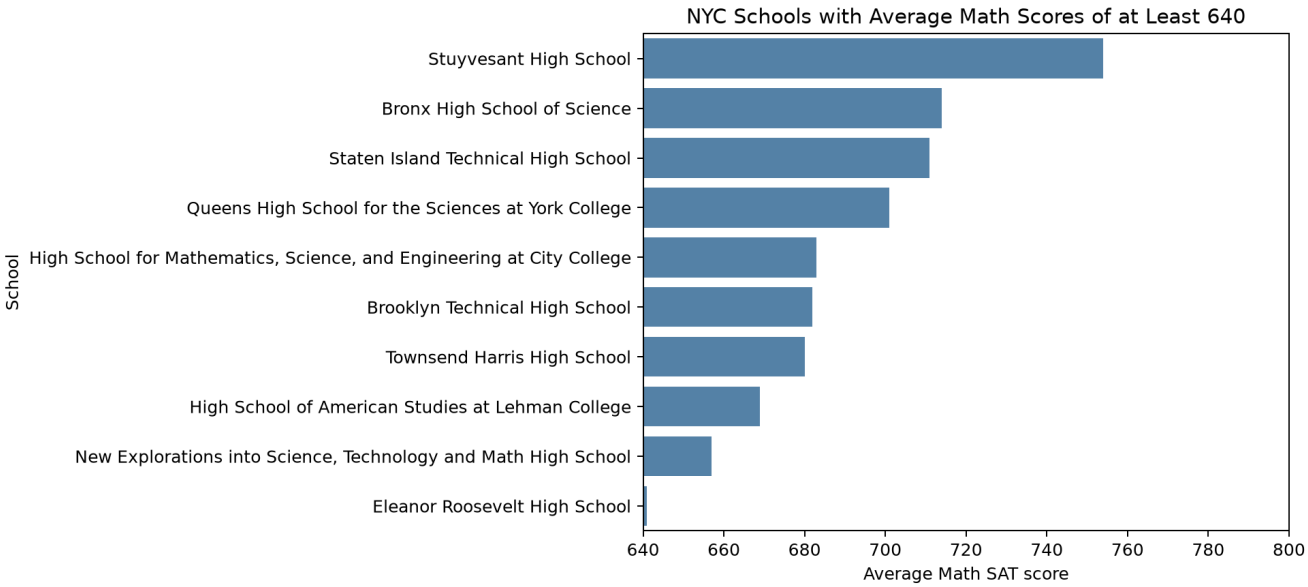
Top 10 schools by combined SAT

School	Borough	Combined SAT
Stuyvesant High School	Manhattan	2,144
Bronx High School of Science	Bronx	2,041
Staten Island Technical High School	Staten Island	2,041
High School of American Studies at Lehman College	Bronx	2,013
Townsend Harris High School	Queens	1,981
Queens High School for the Sciences at York College	Queens	1,947
Bard High School Early College	Manhattan	1,914
Brooklyn Technical High School	Brooklyn	1,896
Eleanor Roosevelt High School	Manhattan	1,889
High School for Mathematics, Science, and Engineering at City College	Manhattan	1,889

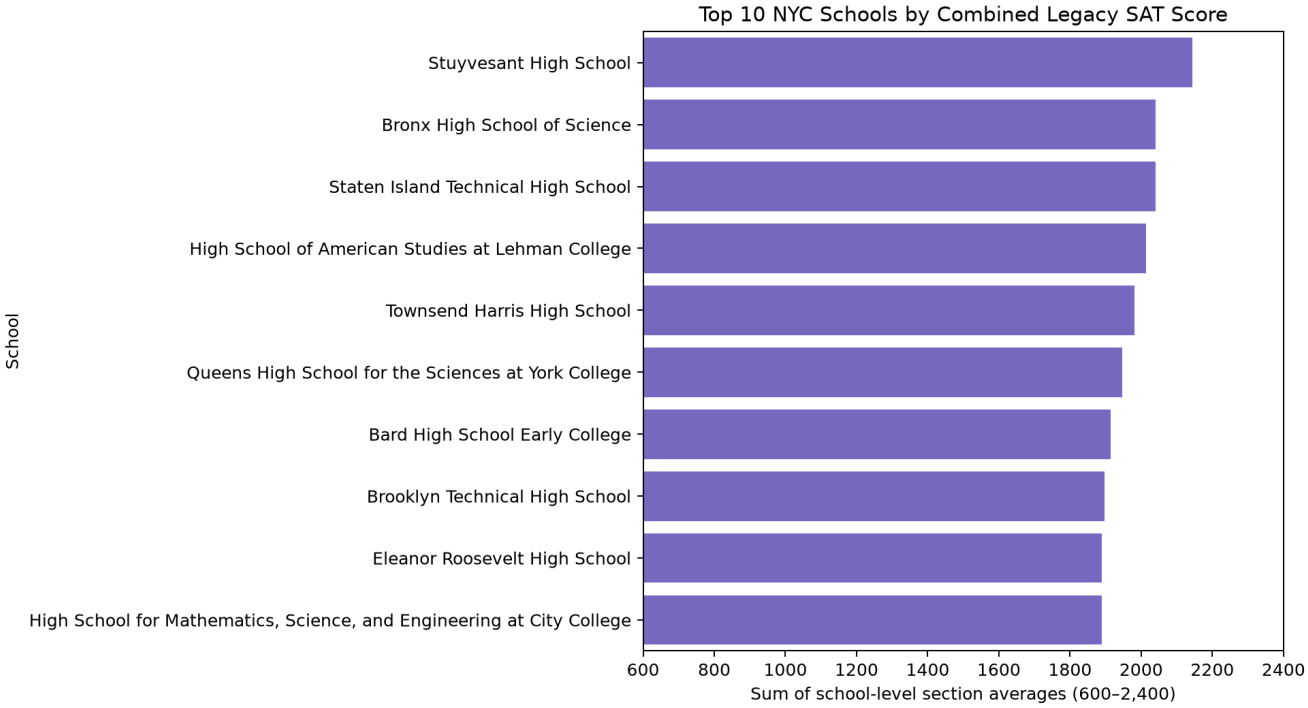
Borough summary (unweighted school-level scores)

Borough	Schools	Mean	Median	Sample SD
Bronx	98	1,202.72	1,190.00	150.39
Brooklyn	109	1,230.26	1,186.00	154.87
Manhattan	89	1,340.13	1,284.00	230.29
Queens	69	1,345.48	1,290.00	195.25
Staten Island	10	1,439.00	1,382.00	222.30

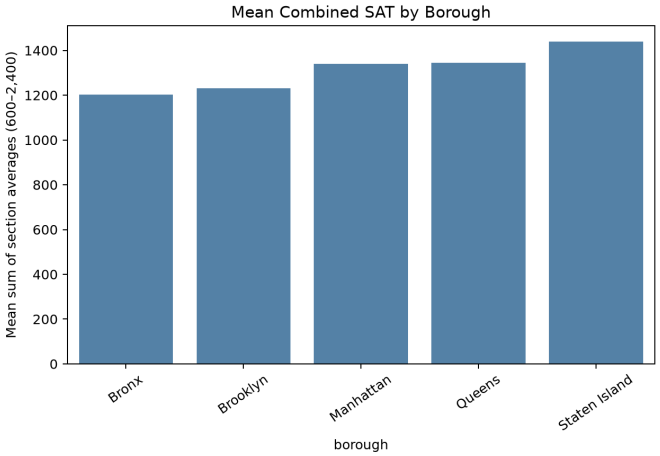
Schools meeting the Math SAT threshold

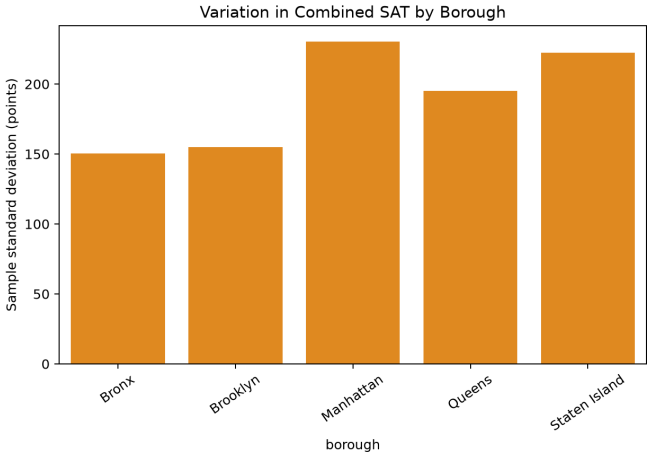


Ten highest combined legacy SAT school averages

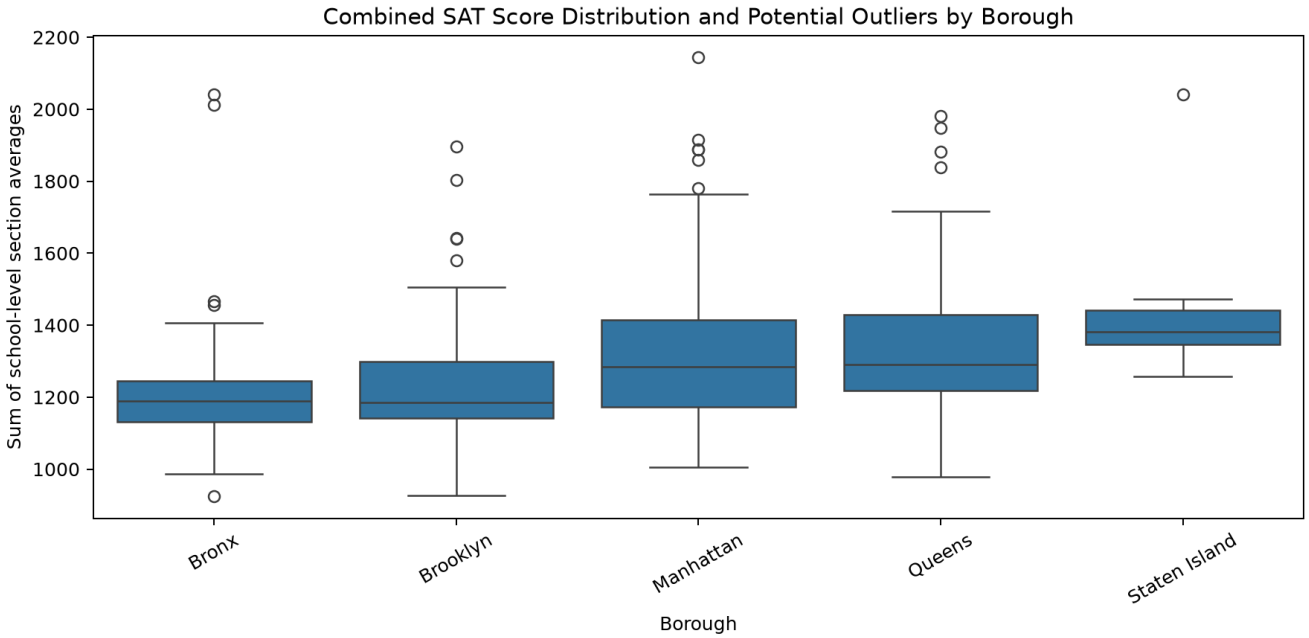


Borough means and within-borough variation

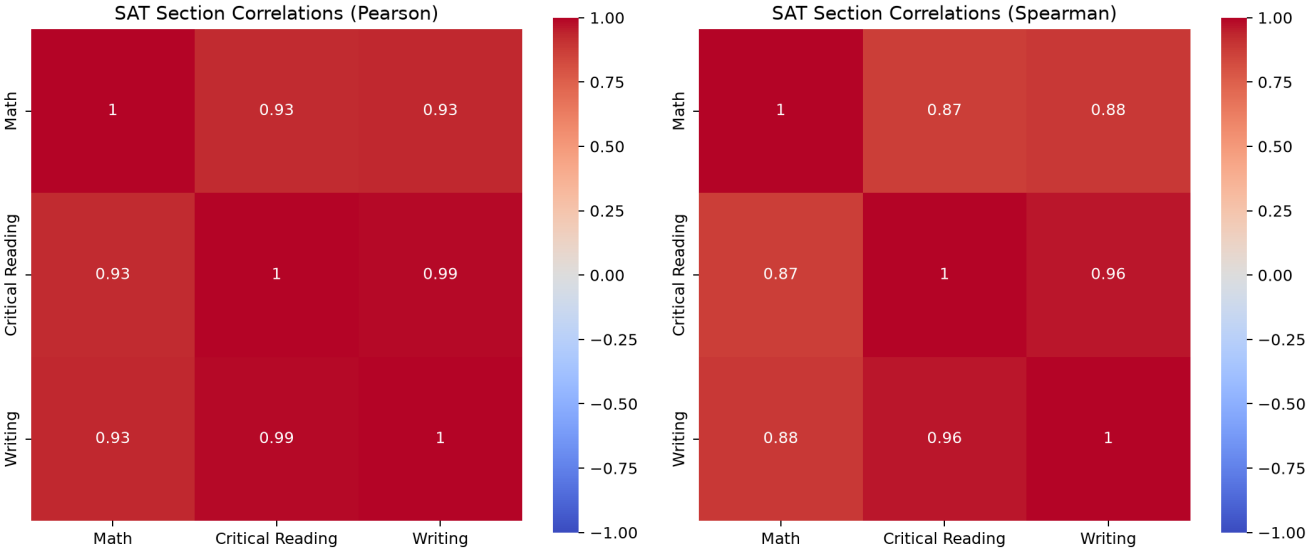

Mean Combined SAT by Borough
Mean sum of section averages (600–2,400)
0
200
400
600
800
1000
1200
1400
Bronx
Brooklyn
Manhattan
Queens
Staten Island
borough


Variation in Combined SAT by Borough
Sample standard deviation (points)
0
50
100
150
200
Bronx
Brooklyn
Manhattan
Queens
Staten Island
borough

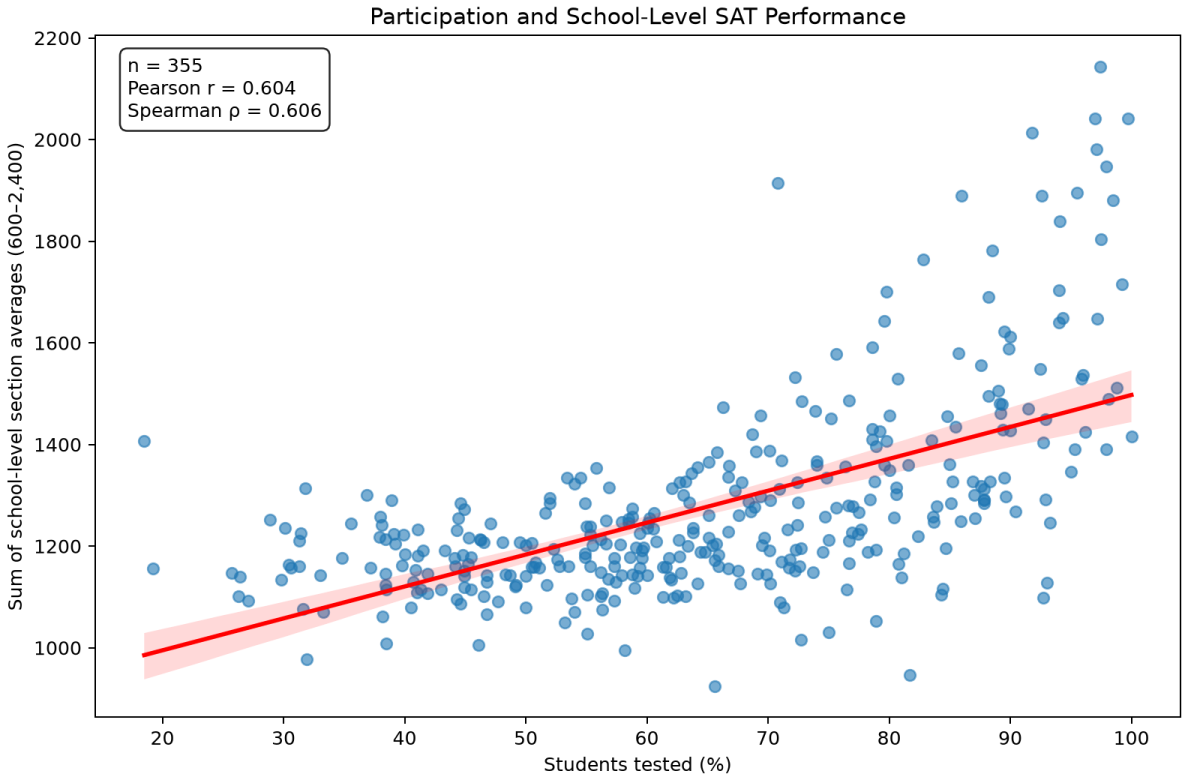
Combined SAT distribution by borough



Pearson and Spearman relationships among SAT sections



Participation versus school-level SAT performance



Interpretation and limitations

Borough means are unweighted: each school contributes equally because the file does not contain enrollment or test-taker counts. They are not student-weighted borough averages. Staten Island has a much smaller school group, so its summary should be interpreted with care.

The standard deviations use the sample formula (ddof=1). A larger standard deviation means greater spread among the recorded schools, not better performance. IQR outliers are prompts for review, not automatic errors.

Correlation describes association and does not establish causation. School-level averages cannot be used to infer individual student outcomes. The participation analysis excludes schools with missing percentages, and missingness may not be random.

SAT averages alone do not measure overall school quality. The file uses the legacy three-section SAT format and its exact testing year is not established here. Results should prompt further investigation, not serve as a sole basis for school rankings or funding decisions.

Reproducibility: the accompanying Jupyter notebook contains the full workflow. Charts in this PDF are exported by executing that notebook from the repository.